



FIT BUDDY 2.0

AI-ASSISTED FITNESS PLATFORM

EXPANDED • VISUAL • TECHNICAL PROJECT REPORT

Flask • Gemini 2.5 Flash • SQLite • Chart.js • html2pdf.js

01 • PROJECT OVERVIEW

Fit Buddy 2.0 combines AI-assisted workout planning with a lightweight web application for coaching conversations, session logging and personal progress views. The central product idea is to turn a user’s stated training target into an actionable routine, then provide a structured place to record completed activity.

The design separates responsibilities: AI generates adaptable content, while application code provides structure, persistence and presentation so model output can be rendered consistently.

CORE CAPABILITIES

PLAN	Goal-oriented workouts with exercises, sets, repetitions and rest guidance.
COACH	Conversational fitness questions and follow-up prompts.
TRACK	Session metadata, history, streaks and progress summaries.
EXPORT	Browser-rendered routines converted into downloadable PDFs.

Scope note: the supplied report describes the coach as a trainer/nutritionist persona; this report treats the output as general informational support, not medical advice.

TECHNOLOGY STACK

Layer	Responsibility	Technology
Presentation	Inputs, cards, charts, PDF export	Browser • Chart.js • html2pdf.js
Application	Routes and response formatting	Flask / app.py
AI	Workout and conversation generation	Gemini 2.5 Flash
Persistence	Profile and workout logs	SQLite / fitbuddy_pro.db

02 • TECHNICAL ARCHITECTURE

The browser handles presentation and client-side visualisation. Flask is the application boundary for API requests, persistence and calls to the external generative-AI service. [filecite turn0file0 L63-L81](#)

SYSTEM ARCHITECTURE

BROWSER UI Inputs • cards Chart.js • html2pdf.js	FLASK Routes Validation • formatting	SERVICES & DATA Gemini 2.5 Flash SQLite
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Typical path: browser request → Flask route → AI service or database → JSON response → interface update. [filecite turn0file0 L81-L89](#)

WORKOUT-GENERATION FLOW

01	Capture intent	User selects or enters a target.
02	API request	POST /api/workout with JSON.
03	Defaults	“Full Body” is used if no target is supplied.
04	Generation	Gemini returns JSON matching the response schema.
05	Response	Endpoint returns status and model output for rendering.

Documented gap: authentication, deployment topology, retry policy, rate limiting and secret-management configuration are not specified in the source. [filecite turn0file0 L105-L105](#)

03 • DATA MODEL & LIFECYCLE

The reported schema contains a profile table and a workout-log table, providing a basic foundation for profile identification and session history. [filecite turn0file0 L113-L116](#)

DATABASE SCHEMA

Table / field	Type / constraint	Purpose
user_profile.id	INTEGER • PRIMARY KEY	Stable internal identifier.
user_profile.name	TEXT • NOT NULL	Display name; absent values are prevented.
user_profile.streak_count	INTEGER • DEFAULT 0	Consecutive-day count; needs timezone/day rules.
workout_logs.id	PRIMARY KEY • auto	Individual log identifier.
workout_logs.title	TEXT • NOT NULL	Routine name for history/export.
workout_logs.duration_min	INTEGER	Session minutes; validate invalid values.
workout_logs.calories_burned	INTEGER	Estimated energy; label clearly as an estimate.

RECOMMENDED DATA REFINEMENTS

LINK	Add user_id foreign key to workout_logs for multiple profiles.
TIME	Add a consistent completion timestamp, commonly UTC.
TRANSACTION	Validate, write the log and update streak consistently.
ANALYTICS	Aggregate logs over clearly defined time ranges.

These are recommended extensions, not claims about fields already implemented. [filecite turn0file0 L141-L153](#)

04 • ANALYTICS & PERFORMANCE

The source report includes a weekly chart but no underlying numeric dataset or formal metric definitions. The visual below is illustrative only. [\[filecite\turn0file0\L156-L161\]](#)

ILLUSTRATIVE WEEKLY DASHBOARD

Mon		20
Tue		35
Wed		45
Thu		30
Fri		60
Sat		75
Sun		90

Illustrative only: example index values are not measured Fit Buddy user data. [\[filecite\turn0file0\L160-L161\]](#)

METRICS TO DEFINE

Workout frequency	Completed sessions per day/week	Separate planned vs completed
Training volume	Commonly sets × reps × load when load is captured.	Listed schema lacks these fields.
Duration	Sum/average session minutes.	Use consistent units.
Streak	Consecutive calendar days with qualifying sessions.	Define timezone and missed-day rules.
Calories	Sum of recorded estimates.	Do not present as precise measurements.

A meaningful dashboard should state period, units, actual-vs-target status and missing-day handling. [\[filecite\turn0file0\L184-L185\]](#)

05 • IMPLEMENTATION PATTERN

The supplied app.py excerpt demonstrates the essential workout-generation pattern: accept input, call Gemini with a schema-constrained request, and return JSON. [filecite\turn0file0\L203-L219](#)

REPRESENTATIVE BACKEND FLOW

```
@app.route("/api/workout", methods=["POST"]) def generate_workout(): data = request.get_json() or {} target = data.get("target", "Full Body") response = client.models.generate_content( model="gemini-2.5-flash", contents=f"Create workout for target '{target}'", config=types.GenerateContentConfig( response_mime_type="application/json", response_schema={ ... } ) ) return jsonify({"status": "success", "data": response.text})
```

Schema-constrained generation helps the client locate expected fields, but output remains untrusted input and should be parsed and validated before success is returned. [filecite\turn0file0\L220-L224](#)

PRODUCTION HARDENING

INPUT	Validate JSON type, target length and allowed values.
FAILURES	Handle provider timeouts, quota errors and malformed responses.
SECRETS	Keep API keys server-side; avoid secrets and stack traces in responses.
OPERATIONS	Add request limits, safe logging and health monitoring.

The source contains a shortened response_schema={ ... } excerpt; this report explains the pattern rather than presenting a full executable implementation. [filecite\turn0file0\L240-L241](#)

06 • QUALITY ASSURANCE

The original project report marks four test cases as PASS. These are preserved as reported results, not independently executed results. [filecite\turn0file0\L243-L248](#)

REPORTED TEST COVERAGE

TC-01	Workout AI	POST /api/workout	Structured routine for UI cards.	PASS (reported)
TC-02	AI Coach	POST /api/chat	Natural-language coaching reply.	PASS (reported)
TC-03	Streak DB	POST /api/log-workout	Persist log and increment streak.	PASS (reported)
TC-04	PDF export	Client-side generation	Clean html2pdf.js document.	PASS (reported)

HIGH-VALUE FOLLOW-UP TESTS

Schema & validation	Missing target, wrong JSON types, empty response, incomplete exercise fields.
External service	Timeouts, provider errors, quota limits and retry behaviour.
Persistence	Repeated logs, database errors, streak boundaries and timezone transitions.
Export quality	Long routines, page breaks, special characters and mobile layouts.
Security	Unauthorised access, sanitisation, secret exposure and abuse/rate limits.

Status note: the supplied report states “Production Ready,” but this document cannot independently verify deployment readiness, test execution or operational controls. [filecite\turn0file0\L278-L279](#)

07 • CONCLUSION & NEXT STEPS

Fit Buddy 2.0 presents a practical flow: structured AI-generated plans, a Flask service boundary, SQLite-backed activity records, browser analytics and downloadable routines.

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RECOMMENDED NEXT STEPS

01	Define analytics Document formulas, units, periods and missing-data rules.
02	Enrich history Capture fields needed for meaningful volume and progression analysis.
03	Strengthen validation Validate client input and model responses before persistence/rendering.
04	Harden failures Add timeout, quota, malformed-response and database failure paths.
05	Document deployment Record authentication, secret handling, rate limits, topology and monitoring.

DESIGN PHILOSOPHY

AI creates adaptability. Application code creates reliability. Data creates continuity. Visualisation creates understanding. Testing creates confidence.

Prepared as an expanded, visually redesigned edition of the supplied Fit Buddy 2.0 report; recommendations and illustrative material are explicitly separated from source claims. filecite turn0file0 L285-L285